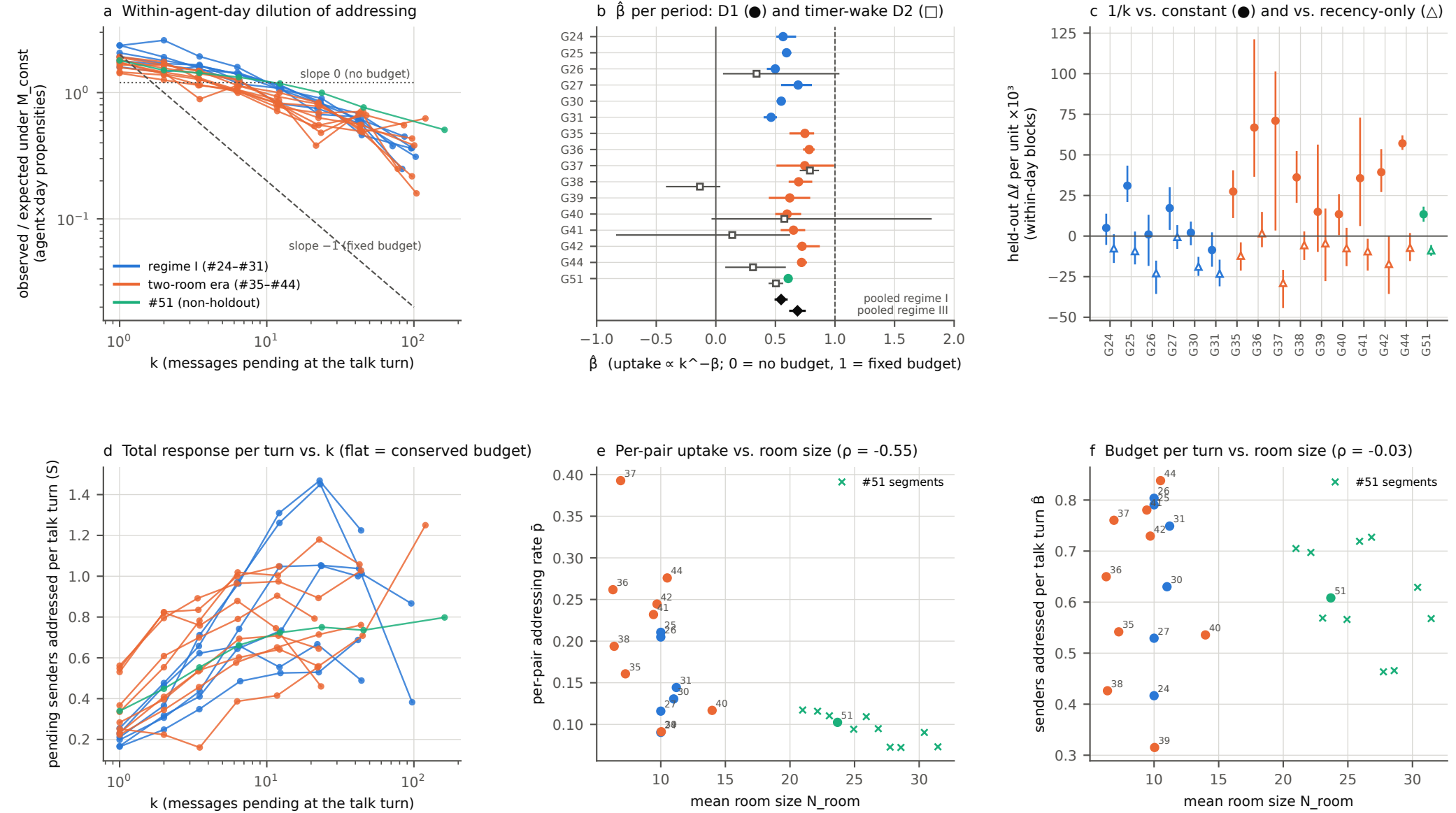


H18 · Attention dilution in the AI Village: exploratory round 1 (non-holdout, 16 goal periods)



P1 dilution: $\hat{\beta}$ CI excludes 0 in 16/16 periods; median $\hat{\beta}$ 0.64; pooled regime I 0.55 ± 0.02 , regime III 0.69 ± 0.03 (I^2 0.92).

P2 shape: near-literal budget (M_{inv} or M_{sat} with $k_0 < 3$) wins CV in 5/16; saturating ($k_0 \approx 3-8$) 6; recency-only 5; constant 0. 1/k beats recency-only in 0/16 (recency beats 1/k, CI: 8).

P3 Σ : pooled ϵ_S 0.32 ± 0.03 . P4 mentions: pooled e^{γ} 2.2; $\beta_M - \beta_{other}$ -0.06 ± 0.07 .

P5 timer wakes (exogenous k): pooled $\hat{\beta}_{D2}$ 0.37 ± 0.14 vs D1 regime III 0.69.

Post hoc: k adds held-out value at fixed recency rank in 5/16 periods (regime I; #51 with $\hat{p} = 1.00$, $k_0 \approx 7$); engagement (i addressed j last turn) does not absorb $\hat{\beta}$.

Placebo fails 16/16: senders of messages the agent could not yet see are mentioned far above baseline (mentions partly mark ongoing exchanges).

Synthetic (axis F): $\hat{\beta}$ recovered ($1/k \rightarrow 1.00 \pm 0.03$; const $\rightarrow 0.00 \pm 0.04$); reactive-timing constant agents fake $\hat{\beta}_{D1} \approx 0.78$ with recency winning, $D2 \approx 0$.